Row-major and Column-major Arrangement of matrix

When a matrix gets stored in memory, all elements of it are stored linearly since computer's memory can only be viewed as consecutive units of memory. This leads to two possible arrangements of elements in memory-

- Row-major Arrangement
- Column-major Arrangement

```
int a[3][4] = {
        {12, 1, -9, 23},
        {14, 7, 11, 121},
        {6, 78, 15, 34},
    };
```

Row Major Arrangement:-

← 0th Row →				← 1st Row →				← 2nd row →			
12	1	-9	23	14	7	11	121	6	78	15	34
502	506	510	514	518	522	526	530	534	538	542	546

Column Major Arrangement:-

← 0th column →			← 1st column →			← 2nd col →			← 3rd col →		
12	14	6	1	7	78	-9	11	15	23	121	34
502	506	510	514	518	522	526	530	534	538	542	546

Row Major

For an array a[m][n], the address of element a[i][j]

$$= \text{Base address} + (i * n + j) * w$$

m = 3
n = 4
i = 1
j = 3

Eg: Address of element $a[1][3] = 502 + (1 * 4 + 3) * w$
$= 502 + 7 * 4$
$= 530$

Col. major

$$= \text{Base address} + (j * m + i) * w$$

$$= 502 + (3 * 3 + 1) * 4$$

$$= 502 + 40 = 542.$$

Where, w is widthof the data type example, w=1 for char type, w=4 for float type, etc.